Andrew Y. Guo

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Education

University of Toronto 2026 – 2028

Doctor of Philosophy - Computer Science

- Directly transitioned from MSc, *advisors:* Prof. David Lindell, Prof. Kyros Kutulakos
- Focus on 3D computer vision, neural rendering, agentic AI

University of Toronto 2024 – 2026

Master of Science - Computer Science

- Research stream, *advisors:* Prof. David Lindell, Prof. Kyros Kutulakos
- Masters thesis published at CVPR 2026 as Highlight (Dark3R), Grade: 4.0/4.0

University of Waterloo 2019 – 2024

Honours Bachelor of Software Engineering

- Dean's Honour List, Cumulative Avg: 92%
- Completed 40 months of full-time work experience in industry, double the requirement for co-op program

Publications

Dark3R: Learning Structure from Motion in the Dark

***Andrew Y. Guo**, Anagh Malik, SaiKiran Tedla, ..., Sotiris Nousias, Kiriakos N. Kutulakos, David B. Lindell*

CVPR 2026 (Highlight - Top 5% of Submissions) - **Paper Link** - **Project Page**

BOP Challenge 2024 on Model-Based and Model-Free 6D Object Pose Estimation

*Van Nguyen Nguyen, Stephen Tyree, **Andrew Y. Guo**, ..., Yann Labbe, Martin Sundermeyer, Tomas Hodan*

CVPR CV4MR Workshop 2025 (Best Paper Award) - **Paper Link** - **Project Page**

HANDAL: A Dataset of Real-World Manipulable Object Categories with Pose Annotations, Affordances, and Reconstructions

***Andrew Y. Guo**, Bowen Wen, Jianhe Yuan, Jonathan Tremblay, Stephen Tyree, Jeffrey Smith, Stan Birchfield*

IROS 2023 - **Paper Link** - **Project Page**

Experience

Research Engineering Intern Sept. 2022 – Apr. 2024

Nvidia *Toronto, ON*

- Researched **3D reconstruction**, **neural scene representations**, **LiDAR novel view synthesis**, and **pose estimation**
- Created automated **6D pose** annotation pipeline for data collection in the wild, **conference paper** in **IROS 2023**
- *Advisors:* Dr. Stan Birchfield, Dr. Liila Torabi, Dr. Hammad Mazhar

Simulation Software Engineering Intern May 2022 – Aug. 2022

Nvidia *Toronto, ON*

- Designed **synthetic data generation** pipeline for 6D Pose Estimation, **beat benchmarks** in original paper
- Created custom annotator to capture semantic information from simulation to generate ground truth for training
- Sped up data generation by **150%** by asynchronously writing to **s3** bucket with multiprocessing

Machine Learning Intern Sept. 2021 – Dec. 2021

Nvidia *Santa Clara, CA*

- Created features for driver-safety system: **developer.nvidia.com/drive/drive-ix**
- Optimized 11 **convolutional neural networks** (CNNs) using **TensorRT** to run inference on limited hardware
- Transitioned team's entire **SDK** to new version of Nvidia's DRIVE OS to support integration of new camera

Software Engineering Intern May 2021 – Aug. 2021

Canoo *Torrance, CA*

- Created integrated calibration framework in **C** to update ECU calibration without recompiling main application

- **Saved** company **$30,000** in annual licensing fees by creating **Python** module to generate .a2l description files

Autonomous Vehicle Software Intern Jan. 2021 – Apr. 2021
WATonomous *Waterloo, ON*
- Created localization pipeline to match camera and **LiDAR** images with *a priori* map to navigate car without GPS
- Achieved **3x** speed up using multi-processing in C++ with **OpenMP** and lane line matching with **KD-trees**
- Utilized Carla for autonomous driving simulations when access to vehicle was not possible

Software Engineering Intern May 2020 – Aug. 2020
AMD *Toronto, ON*
- Automated analysis of memory dumps and sped up debugging by detecting anomalies and key metrics
- Created register save/restore tools that interfaced with low-level drivers for power-saving features on graphics chips

Teaching Experience

CSC420: Introduction to Image Understanding Fall 2025
Teaching Assistant *University of Toronto*

CSC111: Foundations of Computer Science II Winter 2025
Teaching Assistant *University of Toronto*

SE464: Software Design and Architectures Fall 2023, Winter 2024
Teaching Assistant and Course Curriculum Designer *University of Waterloo*

Awards and Scholarships

Queen Elizabeth II Graduate Scholarship in Science & Technology 2025
- $15,000 Scholarship from the University of Toronto

Sandford Fleming Foundation Exchange Scholarship 2023
- $2000 Scholarship from the University of Waterloo

American Bureau of Shipping Scholarship 2022
- $8000 Scholarship from the University of Waterloo

President's Research Award 2022, 2021
- $1500 Scholarship from the University of Waterloo for Undergraduate Research
- Award twice in consecutive years

Mensa Canada Frank & Betty Woodhams Memorial Scholarship 2021
- $5000 Scholarship from Mensa Canada

University of Waterloo Entrance Scholarship 2019
- $15,000 Entrance Scholarship

President's Scholarship of Distinction 2019
- $2000 Entrance Scholarship from the University of Waterloo

BC Provincial Scholarships 2019
- Awarded $5000 in scholarships from the Government of British Columbia for undergraduate studies

Loblaws National Scholarship 2019
- $1500 Scholarship from Loblaws for undergraduate studies

Volunteering

Project Developer May 2021 – Apr. 2022
UW Blueprint
- Used **React, TypeScript, GraphQL** and **MySQL** to build a translation platform for PlanetREAD, a NPO in India with **500,000 users** aiming to improve literacy by removing language barriers
- Architected database models for storing hundreds of translations, users, and stories
- **Open source** full-stack project, code can be found at: **github.com/uwblueprint/planet-read**